RF3172/6/8

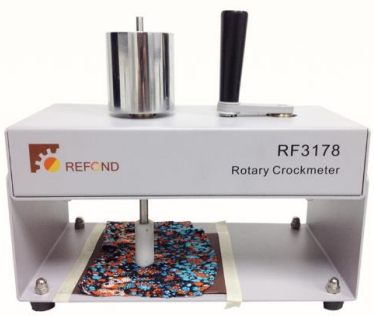
Crockmeter

INSTRUCTION MANUAL



RF3172





RF3178

Version 1.21

REFOND

www.refondtex.com

GENERAL EQUIPMENT SPECIFICATIONS

This page is a record of your equipment specifications. This information is found on the stamped nameplate of your instrument. Please fill in the blanks below when you receive your REFOND instrument.

When contacting the sales or service department to order parts or obtain information, refer to this page. This will allow us to respond quickly and accurately to your request.

MODEL NO. _____

SERIAL NO. _____

230VAC	50/60HZ	1
VOLTS	CYCLES	PHASE

NUMBER	REV. NO.	DATE

MAIN FUSE. _____

SINGLE PHASE_____AMPERES

MODEL: RF3172/6/8
TYPE: Crockmeter

SAFETY INSTRUCTIONS
READ THESE INSTRUCTIONS BEFORE USE

Due to the potential hazards associated with any electrical equipment it is important that the user is familiar with the instructions covering the capabilities of, and the operation of this instrument. The user should ensure that all reasonable safety precautions are followed and if any doubt exists, should seek professional advice before proceeding.

This instrument is designed for use by suitably trained, competent personnel in a controlled working area, and is intended to be used as a bursting strength tester only.

This instrument have moving parts and while all reasonable steps have been taken to protect personnel associated with these moving parts, incorrect or mis-use of the equipment could result in injury. Consideration should be given to the nature of these moving parts when installing and setting up the equipment. This equipment should be installed by fully trained, competent personnel.

REFOND cannot be held responsible for any unauthorized modifications to the unit.

WARNING

This unit contains moving parts and hazardous live voltages. Under no circumstances should the user try to prevent / restrict the movement of the arm or gain access to the internal circuitry, personally or with the aid of a foreign body.

PROVISION FOR LIFTING AND CARRYING

When unpacking or moving this instrument, extreme care is required owing to its physical construction and weight.

It is recommended that accepted lifting and carrying procedures are employed and that personnel wear the appropriate protective equipment e.g. safety shoes. If the unit is to be moved an appreciable distance/height, it is recommended that it is moved via a suitable vehicle e.g. a fork lift truck.

EQUIPMENT WARRANTY

REFOND warrants for twelve (12) months from first use or fifteen (15) months from date of shipment, whichever occurs first, that this REFOND instrument has passed our inspection and tests; that it is free from defects in materials and workmanship; and that it will perform according to the applicable published REFOND specifications if installed and operated according to our instructions. This warranty is in lieu of any other warranty expressed or implied. In no event will REFOND be liable for special or consequential damages as a result of any alleged breach of this warranty provision.

The liability of REFOND hereunder shall be limited to repairing or exchanging, at its option, any defective part FOB, REFOND plant. Equipment or parts that have been subjected to abuse, misuse, accident, alteration, neglect, unauthorized repair or installation are not covered by warranty. REFOND shall have the right of final determination as to the existence and cause of the defect.

When parts are repaired or exchanged, the warranty on said part shall continue in effect for the remainder of the original equipment warranty period or for six (6) months following the date of shipment by REFOND, whichever period is longer.

REFOND reserves the right to make changes and improvements in its products without obligation to install these changes in products theretofore manufactured.

PARTS WARRANTY

Parts and material supplied by REFOND are inspected and tested to ensure that they are free from defects and will operate properly when correctly installed and operated in equipment produced by REFOND. Parts (excluding consumable components such as light sources, optical components, diaphragms, cloth, etc.) are under warranty for a period of six (6) months from date of shipment. This warranty is in lieu of any other warranty expressed or implied. In no event will REFOND be liable for special or alleged breach of this warranty provision.

The liability of REFOND hereunder shall be limited to repairing or exchanging, at its option, any defective part FOB, REFOND plant. Parts that have been subject to abuse, misuse, accident, alteration, neglect, unauthorized repair or installation are not covered by warranty. REFOND shall have the right of final determination as to the existence and cause of the defect.

REFOND reserves the right to make changes and improvements in its products without obligation to install these changes in products theretofore manufactured.

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1.0 INTRODUCTION

The Crockmeter tests the transference of color from the surface of one material to another by either wet or dry rubbing. In addition to color transference tests, the Crockmeter performs scuffing, wet/dry abrasion, flexing, powdering, dry-cleaning, ink abrasiveness, and other tests. Test conditions are controlled and reproducible through the use of standard pressure and motion.

Three models are available for automated or manual testing: the traditional, hand-operated RF3172, the motorized RF3176, and the hand-operated, rotary vertical RF3178. Crockmeters meet requirements to perform the tests listed below in addition to other industry and organizational test methods.

Models RF3172 and RF3176:

AATCC Method 8, AATCC Method 165

ASTM F 1319

EN ISO 105 Part X12

ISO 105 Part D02

JIS L 0849, JIS L 1084

Model RF3178:

AATCC Method 116

ISO 105 Part X16

The *Crockmeter Instruction Manual* contains operational procedures and safety precautions. Please read the manual carefully before installing and operating your instrument to ensure its safe use and accurate test results. REFOND strongly recommends reading this manual and cannot be held responsible for operating the Crockmeter in a manner that is inconsistent with the instructions presented.

2.0 SAFETY

This section introduces safety symbols that will appear throughout this manual. Please read and understand all cautions and warnings before using your Crockmeter. In addition, **Section 6.2** provides a list of operational precautions and guidelines to operate the RF3176 motorized model safely.



HIGH VOLTAGE! indicates that ignoring the instructions may lead to electric shock for the operator.



DANGER! indicates that ignoring the instructions may lead to hazardous conditions for the operator.



NOTE! indicates operational hints and useful information.



WARNING! indicates that ignoring the instructions may damage the instrument.

3.0 SPECIFICATIONS

Model RF3172 (Manual)

PHYSICAL

Length: 60 cm (23.6 in)

Width: 16.5 cm (6.5 in)

Height: 20.3 cm (8 in)

Construction: Cold-rolled steel

Finish: Gray baked enamel

Shipping Weight: 6 kg (13 lb)

Model RF3176 (Motorized)

PHYSICAL ENVIRONMENTAL

Length: 60 cm (23.6 in) Indoor use

Width: 22.2 cm (8.75 in) Altitude: up to 2 000 m

Height: 27 cm (10.6 in) Ambient Temperature: 16 to 29 °C (60 to 85 °F)

Relative Humidity: 80% maximum

Construction: Cold-rolled steel Mains Supply Voltage Fluctuations: ±10% of the nominal voltage

Finish: Gray baked enamel Overvoltage Category: II

Pollution Degree: 2

Shipping Weight: 9.5 kg (21 lb)

ELECTRIC

230 V, 50 Hz, MAX. 0.50 A

Model RF3178 (Rotary Vertical)

PHYSICAL

Length: 26 cm (10.3 in)

Width: 15 cm (6 in)

Height: 25.4 cm (10 in)

Construction: Cold-rolled steel

Finish:	Gray baked enamel
Shipping Weight:	5.4 kg (12 lb)
Workspace:	To operate the instrument and perform routine maintenance, approximately 90 x 60 cm (3 x 2 ft) of workspace is recommended for all models.

CE

Meets EU (European Union) requirements for safety. CE stands for Conformité Européenne (European Conformity).

4.0 INSTALLATION

4.1 Unpacking the Instrument

When you receive your Crockmeter, inspect the carton for damage that may have occurred during shipping. Carefully unpack the instrument and thoroughly inspect the components for any damage or shortage. Report any equipment damages to the carrier and/or shortages to REFOND customer service.

4.2 List of Materials

The Crockmeter is shipped with the following items:

- Abrasive paper (two pieces with the RF3172 and RF3176; one piece with the RF3178)
- Spring clip
- Specimen holder (Models RF3172 and RF3176)

4.3 Setup

After unpacking the Crockmeter, place the instrument on a smooth, level surface in a room with steady temperature and humidity. The constant motion of the arm requires a sturdy workspace. (See the product specifications in **Section 3.0**.)

If your instrument is the RF3176, connect the electrical power cord into the receptacle in the back of the instrument. With the POWER button in the OFF position, plug the cord into the proper power outlet.

5.0 RF3172 OPERATION

The RF3172 is the hand-operated Crockmeter and is ideal for short-run tests. It has a resettable 0 to 999-stroke counter that counts the number of times the meter handle is cranked. Before performing a test, the operator covers the instrument's base with abrasive paper to keep the test specimen from sliding. The specimen is then clamped atop the paper to the base with the specimen holder. Test cloth is fastened to the rubbing finger with a spring clip. Beginning with the finger resting on the specimen at the base's front end, the operator cranks the meter handle for the specified number of strokes.



The finger, which rests on the specimen with a force of 9 newton, traverses a pre-set path of 104 +/- 3 mm (4 in). The stroke length can be adjusted to 50 or 76 mm (2 or 3 in) depending on the specimen's size. The test cloth is removed after the specified number of strokes and the circular image is evaluated according to the test method.

5.1 Mounting the Specimen and Test Cloth

1. Attach the rubbing arm to the hook on the drive arm above so that the finger no longer rests on the base of the instrument.
2. Center a piece of abrasive paper, abrasive side up, on the base of the Crockmeter. The paper's long dimension should be in the direction of the rubbing. With the paper's short edge meeting the base's front edge, tape the paper's long sides to the base.
3. Place the specimen flat and face-up on the abrasive paper. The specimen's long dimension should be in the direction of the rubbing with the finger centered and resting on test area.
4. To secure the specimen, center the specimen holder over the instrument's base with the holder's enclosed end flush with the base's front edge. Push

the holder down firmly over the base until it lies directly on the specimen.

5. Center a piece of test cloth over the finger so that the weave is parallel with the direction of rubbing. Wrap the sides of the cloth evenly around the finger, making sure the cloth is smooth and taut over the finger's rubbing area.
6. With the spring clip's loops facing upward, secure the cloth by squeezing the loops together and pushing the spring clip onto the finger. Lower the covered finger onto the test specimen.



WARNING! To keep from damaging the finger or arm, do not allow the arm to drop onto the instrument's base.

5.2 Adjusting the Stroke Length

The path, or stroke length, traversed by the rubbing finger may be adjusted to 50, 76 or 104 mm (2, 3, or 4 in) to accommodate the specimen's size. To adjust the stroke length, use a flat screwdriver to loosen the screw securing the drive arm to the crank arm. Note the orientation of the washer for the proper spacing between the arms. Move the drive arm to the desired setting and tighten the screw.

5.3 Resetting the Cycle Counter

The cycle counter is located on the shaft between the crank handle and drive arm. To reset the counter, turn the knob counterclockwise until the counter reads "000."

5.4 Testing a Specimen

1. Make sure your specimen and test cloth are properly mounted (see **Section 5.1** above).
2. Reset the cycle counter.
3. Position the finger on the specimen at the base's front end and turn the handle at the rate of one turn per second for the number of strokes specified by the test method. (Each back and forth motion of the arm equals one stroke. For accuracy, note the change on the stroke counter.)
4. When the desired number of strokes is reached, remove the specimen and evaluate it according to the test criteria.

6.0 RF3176 OPERATION

The RF3176 is the motorized Crockmeter and is recommended for frequent use and long-run tests. This model provides a uniform stroke rate of 60 revolutions per minute, automatic shut-off, a digital cycle counter, and a removable stroke arm safety cover.

Before performing a test, the operator covers the instrument's base beneath the finger with abrasive paper to keep the specimen from sliding. A test specimen is then clamped atop the paper to the base with the specimen holder. Test cloth is fastened to the rubbing finger with a spring clip. The RF3176 uses a digital cycle counter that automatically counts up to the specified number of strokes and stops, sounding a buzzer. The finger, which rests on the specimen with a force of 9 newton, traverses a pre-set path of 104 +/- 3 mm (4 in).



The stroke length can be adjusted to 50 or 76 mm (2 or 3 in) depending on the specimen's size. The test cloth is removed after the specified number of strokes and the circular image is evaluated according to the test method.

6.1 Operational Precautions

6.1.1 Emergency Power Disconnect

In an emergency, unplugging the cord from the RF3176's back panel socket will disconnect all power to the instrument. The RF3176 will immediately shut down and the test will stop. To resume operation, turn the power switch in front of the instrument to OFF, connect the cord, and turn the power switch to ON.

6.1.2 Stopping a Test in an Emergency

Turn off the circuit breaker ON/OFF switch beside the cycle counter and the RF3176 will shut off completely.

The counter will display the counted value; Turn on the circuit breaker ON/OFF

switch and push the start button, the test can be continued & the counter is continued count up.

6.1.3 Fuse Replacement

The RF3176's cycle counter is protected against overload by a glass cartridge fuse. The fuse is easily accessible on the instrument's central panel on the back of the RF3176 where the power cord enters the housing. Replace it only with a 0.25 A/250 V, slow blow fuse.



If the fuse blows or is removed, all instrument components power will be shut off and the arm will stop.



Never operate the RF3176 without the protective base cover installed over the linear reciprocating arm. Never touch the moving parts of the RF3176 while it is operating.



Use caution when working with liquids and electricity because of a potential electrical shock hazard.

6.2 Mounting the Specimen and Test Cloth

1. Hold the rubbing arm up so that the finger no longer rests on the base of the instrument and move the hold-up pin to the right to support the rubbing arm.
2. Center a piece of abrasive paper, abrasive side up, on the base of the Crockmeter. The paper's long dimension should be in the direction of the rubbing. With the paper's short edge meeting the base's front edge, tape the paper's long sides to the base.
3. Place the specimen flat on the abrasive paper, image side up. The specimen's long dimension should be in the direction of the rubbing with the finger centered and resting on test area.
4. To secure the specimen, center the specimen holder over the instrument's base with the holder's enclosed end flush with the base's front edge. Push

the holder down firmly over the base until it lies directly on the specimen.

5. Center a piece of test cloth over the finger so that the weave is parallel with the direction of rubbing. Wrap the sides of the cloth evenly around the finger, making sure the cloth is smooth and taut over the finger's rubbing area.
6. With the spring clip's loops facing upward, secure the cloth by squeezing the loops together and pushing the spring clip onto the finger. While holding the rubbing arm up, move the holdup pin to the left to release the arm for operation. Lower the covered finger onto the test specimen.



To keep from damaging the finger or arm, do not allow the arm to drop onto the instrument's base.

6.3 Adjusting the Stroke Length

The path, or stroke length, traversed by the rubbing finger may be adjusted to 50, 76 or 104 mm (2, 3, or 4 in) to accommodate the specimen's size. To adjust the stroke length, follow the steps below.

1. Unplug the RF3176 from its power source.
2. Loosen the thumb screws securing the protective base cover over the linear reciprocating arm and lift off the cover. It is not necessary to remove the screws.
3. Use a flat screwdriver to loosen the screw securing the drive arm to the crank arm. Note the orientation of the washer for the proper spacing between the arms. Move the drive arm to the desired setting and tighten the screw.
4. Reinstall the protective base cover and tighten the thumb screws.

6.4 Setting and Resetting the Digital Cycle Counter

To operate the RF3176, the cycle counter must first be set to the desired number of strokes. The START button is then pressed to begin the test. When the counter counts up to the set point, the RF3176 shuts off immediately and the buzzer sounds. To set the counter, follow the procedure below:

1. Press the arrow key  until the display digit you wish to change is flashing.

2. Press the arrow key  or key  until the desired digit appears.



3. Press the arrow key  to select another digit for alteration and repeat **Step 2**.

4. Press Steps 2 and 3 until the counter displays the desired cycles for you test.

5. Press key  to return to RUN mode.

6. When the counter counts up to the set point and the buzzer sounds, press the reset key  can stop the sounds.

6.5 Testing a Specimen

1. Make sure your specimen and test cloth are properly mounted (see **Section 6.0**).
2. Make sure the protective base cover is secured in place.
3. Set the cycle counter as described in **Section 6.4**. Press the START button to initiate the test.
4. When the cycle counter counts up to the selected number of strokes, the buzzer will sound and the test will stop automatically. Meantime, the timer will be reset to the SET value automatically.
5. Remove the specimen for evaluation according to the test criteria.
6. To start a new test, repeat the above steps.

Remarks: When the power is OFF during testing, the counter does not have memory. A new specimen should be tested instead of the current one.

7.0 RF3178 OPERATION



The manually operated RF3178 has a reciprocating rotary finger and is used for tests with prints on small colored areas. When the standard weight is used, the 16-mm diameter finger rests on the specimen with a force of 11.1 newton. The finger rotates 1.125 turns clockwise, then counterclockwise as the operator turns the handle in one direction. The pressure in conjunction with abrasive paper on the base of the instrument holds the specimen immobile while being tested. The head of the instrument tilts away from the base for easy loading and unloading of specimens.

7.1 Mounting the Specimen and Test Cloth

1. Remove the test weight from the vertical rod.
2. Tilt the upper half of the instrument away from you to access the base. Center a piece of abrasive paper, abrasive side up, on the base of the instrument beneath the finger. Tape the paper's short sides to the base.
3. Center a piece of test cloth over the finger and wrap the sides of the cloth evenly around the finger. Make sure the cloth is smooth and taut over the finger's rubbing area. With the spring clip's loops facing upward, secure the cloth by squeezing the loops together and pushing the spring clip onto the finger.
4. Hold a test specimen on the abrasive paper at the point where the finger meets the base. Tilt the upper half of the instrument back to the original position so that the test cloth is in contact with the test specimen.
5. Place the test weight on the vertical rod to apply pressure to the finger.

7.2 Testing a Specimen

1. Holding the test specimen in place with one hand, turn the hand crank at

the rate of one revolution per second for the number of turns specified by the test. (One turn of the crank produces two reciprocal turns of the finger.)

2. When the desired number of strokes is reached, remove the specimen and evaluate it according to the test criteria.

8.0 RF3172 AND RF3176 OPTIONS

REFOND offers a full line of interchangeable crocking fingers and arms to comply with a variety of industry and organizational test specifications. The most commonly used fingers and arms are listed below. For these and other Crockmeter accessories and replacement parts, contact REFOND Customer Service.

8.1 Replacing the Finger

1. Remove the screw located above the finger on the side of the rubbing arm.
2. Pull the finger straight down out of the hole in the underside of the arm.
3. Insert the replacement finger into the hole and tighten the screw. If the finger has a flattened edge, the flat side should face the screw.

8.2 Acrylic Fingers

The standard rubbing finger for the RF3172 and RF3176 is a 16-mm (0.63-in) diameter acrylic rod. A larger acrylic finger is available to provide a 25 mm (1 in) rubbing surface.

8.3 Wet Test Finger

An optional hollow nylon finger is available for wet and solvent crocking tests. The nylon finger has the same dimensions as the standard finger but offers increased chemical resistance. It has a side-loading hole for applying measured amounts of liquid to be evenly bled into the test cloth by four feeder holes. When filled with a solvent such as MEK (methyl ethyl ketone), for example, it can be used to evaluate the degree of curing and sensitivity to abrasion of ultraviolet-cured printing inks. The fingers are secured with a screw on the side of the arm.



NOTE!

To make sure that the finger is properly seated in the arm, run a test using the Crockmeter calibration cloth and check that the image produced is even.

8.4 Crock Block

The crock block provides a rectangular rubbing surface of 19 x 25.4 mm (0.75 x 1 in) and is used for testing textile floor coverings. It is attached to the drive arm by a screw on the side of the arm. The screw can be replaced by a 10-32 x 20 mm Allen-socket cap screw that can be tightened or loosened by hand. A rectangular slip-on sleeve is provided with the crock block to enable the operator to mount test cloth while the block is still attached to the drive arm.



8.5 Yarn Testing Finger

For testing yarns and threads, an optional acrylic test finger is available. The finger is 25 mm (1 in) in diameter and 51 mm (2 in) long. Positioned on its side and held in place by the standard finger, it provides a wider test area and rounded surfaces to eliminate digging. The standard test cloth is used and is fastened by two spring clips that are provided with the finger.

8.6 Lighter-Weight Arms

The standard arm for the RF3172 and RF3176 rests on the specimen with a force of 9 newton. For tests that specify a lighter force, REFOND offers a mid-weight (3 newton) and lightweight arm. (These arms lack the clip for the RF3172's suspension hook to accommodate holes that lighten the weight.)

To change the arm, remove the screw securing the drive arm to the crank arm (note the orientation of the washer) and remove the rubbing arm assembly from the instrument. Insert the new rubbing arm assembly into the guide track. Attach the drive and lever arms by inserting the washer and screw in the desired stroke-length setting then tighten the screw.

9.0 MAINTENANCE

All Crockmeter models are virtually maintenance free. No lubrication is necessary for the RF3176 and RF3178. To ensure long bearing life of the connecting shaft between the drive handle and the drive linkage in the RF3172, apply 2 or 3 drops of 10W motor oil to the oiler hole once a month.

The 2.29 mm (0.09 in) oiler hole is located where the drive handle is mounted onto the connecting shaft.

Be sure to keep the rubbing block, located at the end of the drive arm above the rubbing finger in the RF3172 and RF3176, clean and free of any abrasive dirt or contamination. Keep the RF3176's sensor free of dust by wiping it with a soft cloth as needed.

In general, keep the Crockmeter clean and free from oil and dirt, which may damage small components over time. Replace worn parts as necessary.